

returns in the past 12 months exceeds a certain threshold value. Following [Papailias et al. \(2021\)](#), we calculate the RSM returns using the following equation:

$$R_t^s | P_{t-12,t-1}^s, q = \begin{cases} +r_t^s, & P_{t-12,t-1}^s > q \\ -r_t^s, & P_{t-12,t-1}^s < q \end{cases}, \quad (2)$$

where $P_{t-12,t-1}^s$ denotes the probability of positive returns of the past 12-month period, and q is the threshold value. For simplicity, we select the most intuitive threshold value, $q = 0.5$, indicating that a long position is established when the empirical probability $P_{t-12,t-1}^s$ is greater than 0.5, whereas a short position is established when this probability is smaller than 0.5.

In line with [Moskowitz et al. \(2012\)](#), when investigating the profitability of the trend-following reversal strategies, we implement a volatility scaling approach to weight the portfolio. This approach calculates the portfolio weights, or the position sizes for individual instruments, in a time-varying way. Specifically, we control the position size of each instrument to be inversely proportional to its ex-ante realised volatility σ_t , which is calculated as follows:

$$\sigma_t^2 = 261 \sum_{i=0}^{\infty} (1 - \delta) \delta^i (r_{t-1-i} - \bar{r}_t)^2, \quad (3)$$

where the parameter δ is defined when the centre of mass is equal to 60 days. The benefit of controlling for volatility is that it can lead to more profitable investment strategies, because it takes a crucial role in adjusting the position size of momentum strategies ([Barroso and Santa-Clara, 2015](#); [Kim et al., 2016](#); [Papailias et al., 2021](#)).

To form a portfolio of various instruments, we calculate the trend-following position signals in the same way as in Equations 1 and 2, and allow the portfolio weight for each instrument to be given as a function of its ex-ante realised volatility σ_t^2 . We use a target value for the annual volatility of 40% as in [Moskowitz et al. \(2012\)](#). This aligns our results with the current literature and also mimics a real-world trading scenario with a capital margin of about 5–20%. Then, the TSM and RSM position returns for an asset s are given by: